UCT703: IT PROJECT MANAGEMENT

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3	0	2	4.0

Course objectives: After completion of this course, students will learn the techniques to effectively plan, manage, execute, and control projects within time and cost targets with a focus on Information Technology and Service Sector. Students will also learn agile project management techniques such as Scrum and DevOps.

Project Overview and Feasibility Studies- Identification, Market and Demand Analysis, Project Cost Estimate, Financial Appraisal.

Project Scheduling: Project Scheduling, Introduction to PERT and CPM, Critical Path Calculation, Precedence Relationship, Difference between PERT and CPM, Float Calculation and its importance, Cost reduction by Crashing of activity.

Cost Control and Scheduling: Project Cost Control (PERT/Cost), Resource Scheduling & Resource Leveling.

Project Management Features: Risk Analysis, Project Control, Project Audit and Project Termination.

Agile Project Management: Introduction, Agile Principles, Agile methodologies, Relationship between Agile Scrum, Lean, DevOps and IT Service Management (ITIL).

Scrum: Various terminologies used in Scrum (Sprint, product backlog, sprint backlog, sprint review, retro perspective), various roles (Roles in Scrum), Best practices of Scrum.

DevOps: Overview and its Components, Containerization Using Docker, Managing Source Code and Automating Builds, Automated Testing and Test Driven Development, Continuous Integration, Configuration Management, Continuous Deployment, Automated Monitoring.

Other Agile Methodologies: Introduction to XP, FDD, DSDM, Crystal.

Laboratory work: Getting acquainted with the popular PM tools. Using Function Point calculation tools for estimation, comparing with COCOMO estimates, Implementation of various exercises using PERT, CPM methods, Preparing schedule, resource allocation etc. Using MS Project, Preparing an RMMM Plan for a case study, Preparing Project Plan for a Software Project for Lab Project or case study. Exploring about PMBOK (Project Management Body of Knowledge) and SWEBOK (Software Engineering Body of Knowledge) from related website, Implementation of software project management concepts using related tools and technologies.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

Upon completion of this course, students will be able to:

1. Describe and apply basic concepts related to IT project planning, scope and feasibility.

2. Analyze IT project estimation techniques.
3. Acquire skills to analyze and manage risks, quality assurance, and project control activities.
4. Describe various project management activities such as tracking, project procurement, configuration management, monitoring.

Text Books:

1. Hughes B. and Cotterell M. and Mall R., Software Project Management, Tata McGraw Hill.
2. Mike Cohn, Succeeding with Agile: Software Development Using Scrum.
3. Pressman R., A practitioner"s Guide to Software Engineering, Tata McGraw Hill.

Reference Books:

1. Roman Pichler, Agile Product Management with Scrum.
2. Ken Schwaber, Agile Project Management with Scrum (Microsoft Professional).